

Kubernetes Components

Instructor

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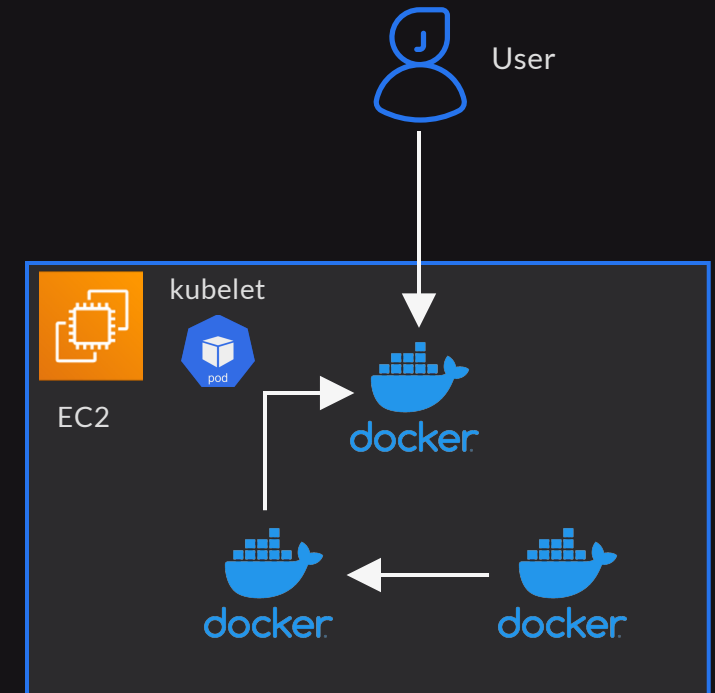
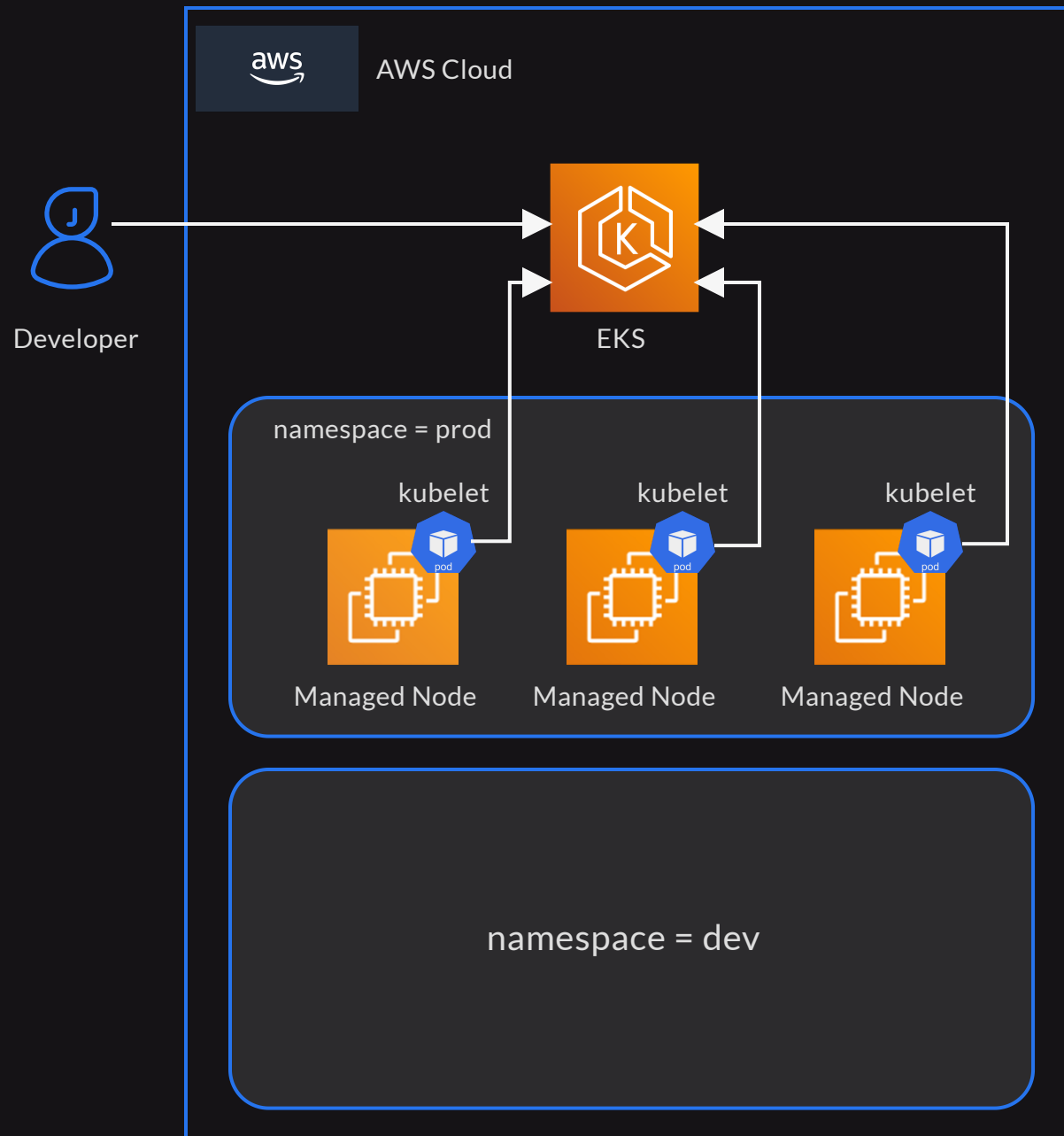
Ex-Head of Engineering - YC-backed startup

Certified AWS Cloud and Kubernetes Engineer

Open source contributor



Managing VMs



Kubernetes

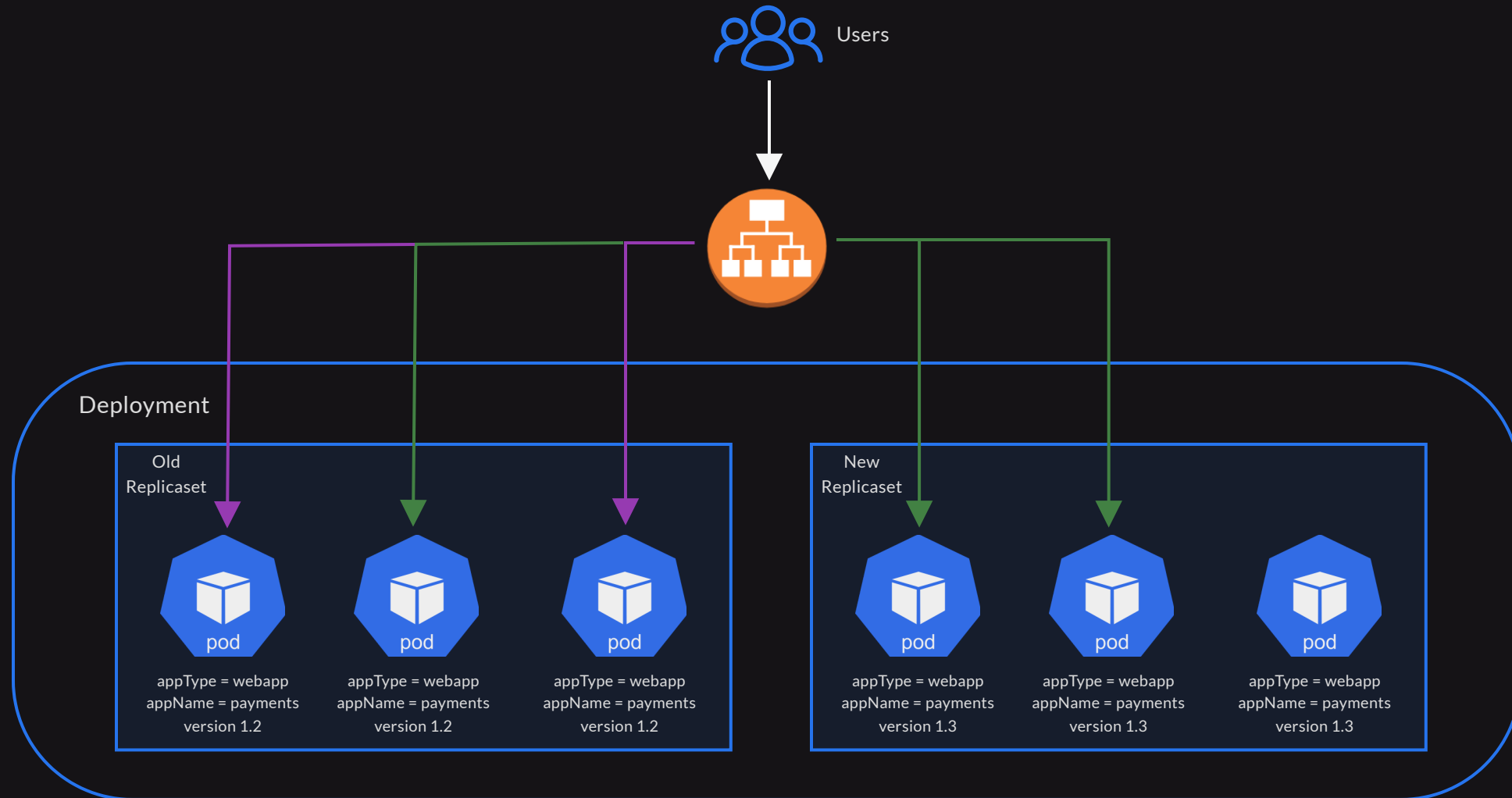
- Pods are the smallest deployable units of computing that you can create and manage in Kubernetes
- A Pod is a group of one or more containers, with shared storage and network resources, and a specification for how to run the containers
- Kubernetes supports many container runtimes such as docker, containerd etc
- Pods can have labels that uniquely defines pod of the same type in a sea of pods
- This can be used to filter pods and manage them
- A pod will be assigned to a node automatically by K8s based on certain conditions



appType = webapp

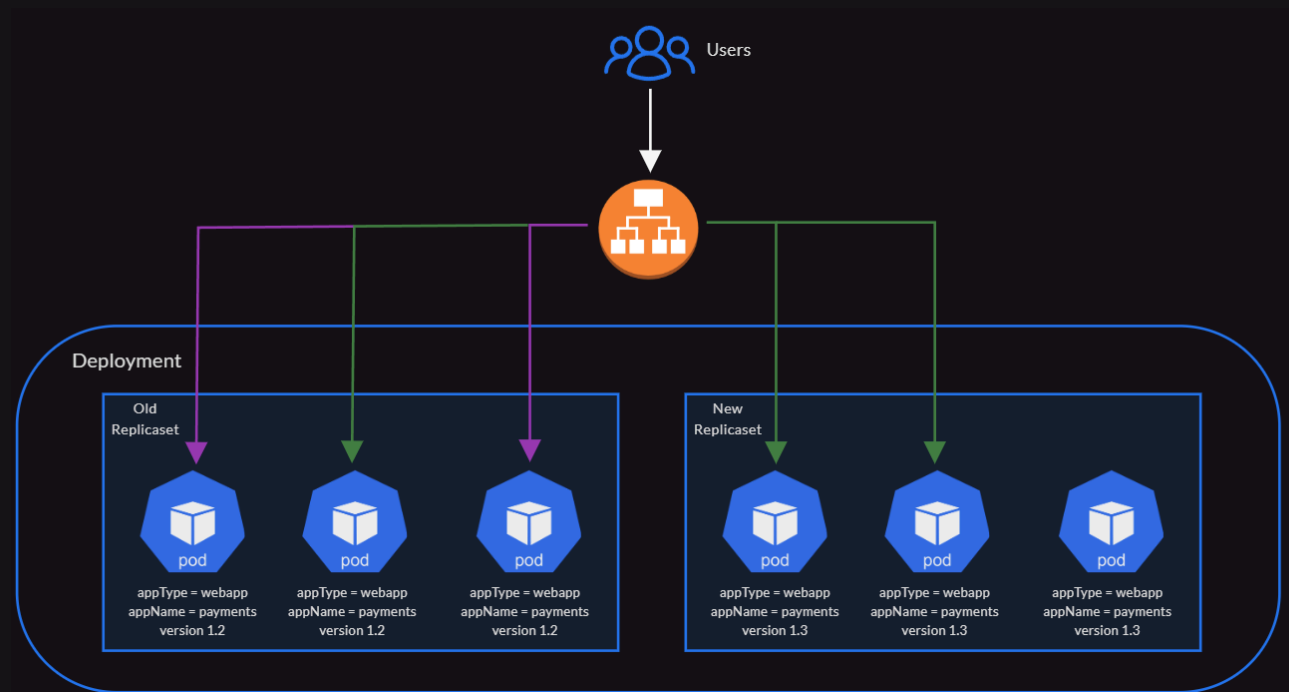
appName = payments

Deployment

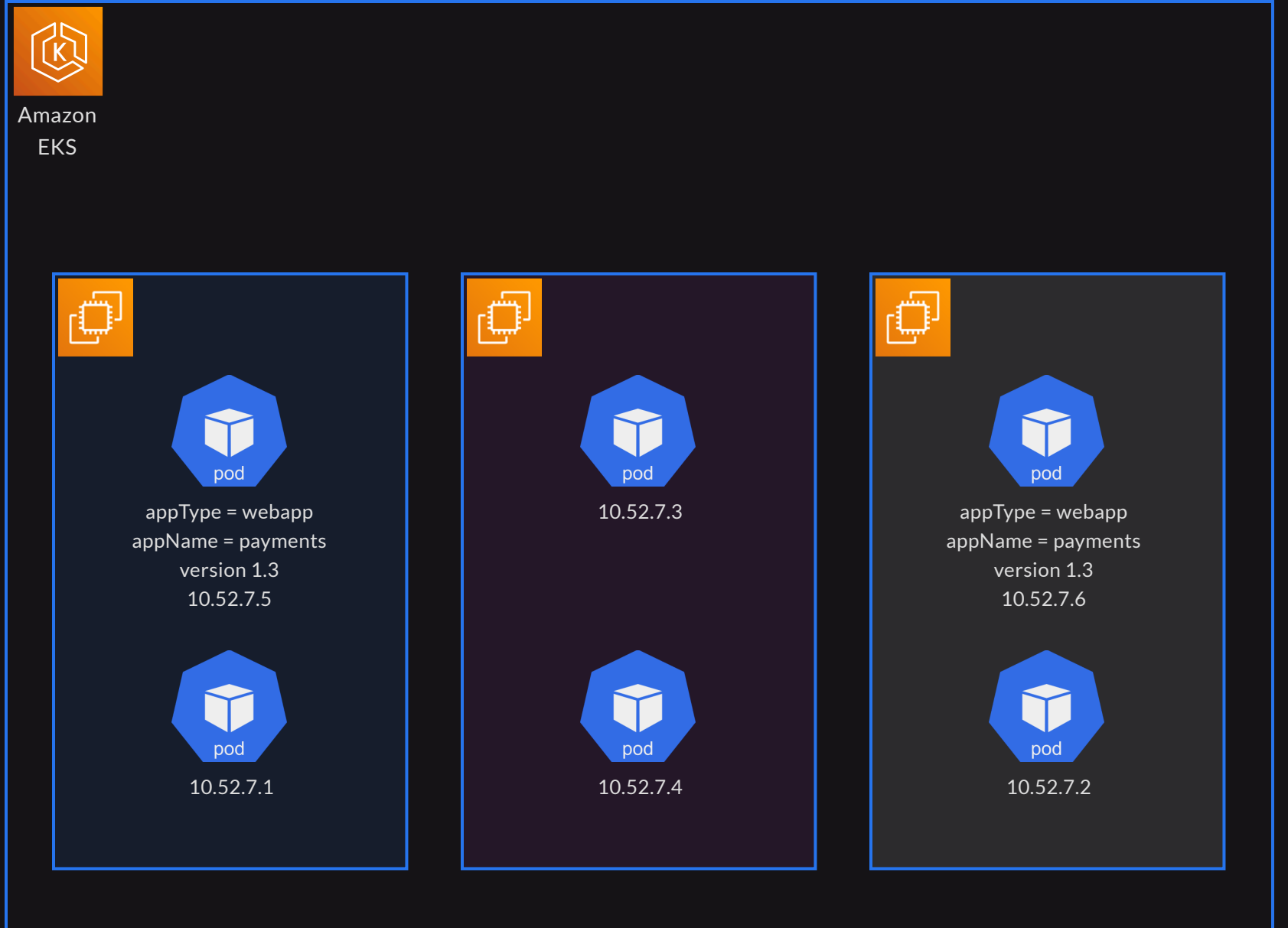


Deployment

- A collection of pods is called a replicaset. This is a logical collection done using **labels** given to a pod and **matchLabels** given to a replicaset
- Health of all the pods in a given replicaset is automatically managed to the desired count
- We can scale a replicaset by a single command
- Deployment manages the lifecycle of an app by controlling its replicaset
- Updating an app version is done by creating new replicaset and deprovisioning old one.
- Types include **rolling** to avoid downtime and **recreate**

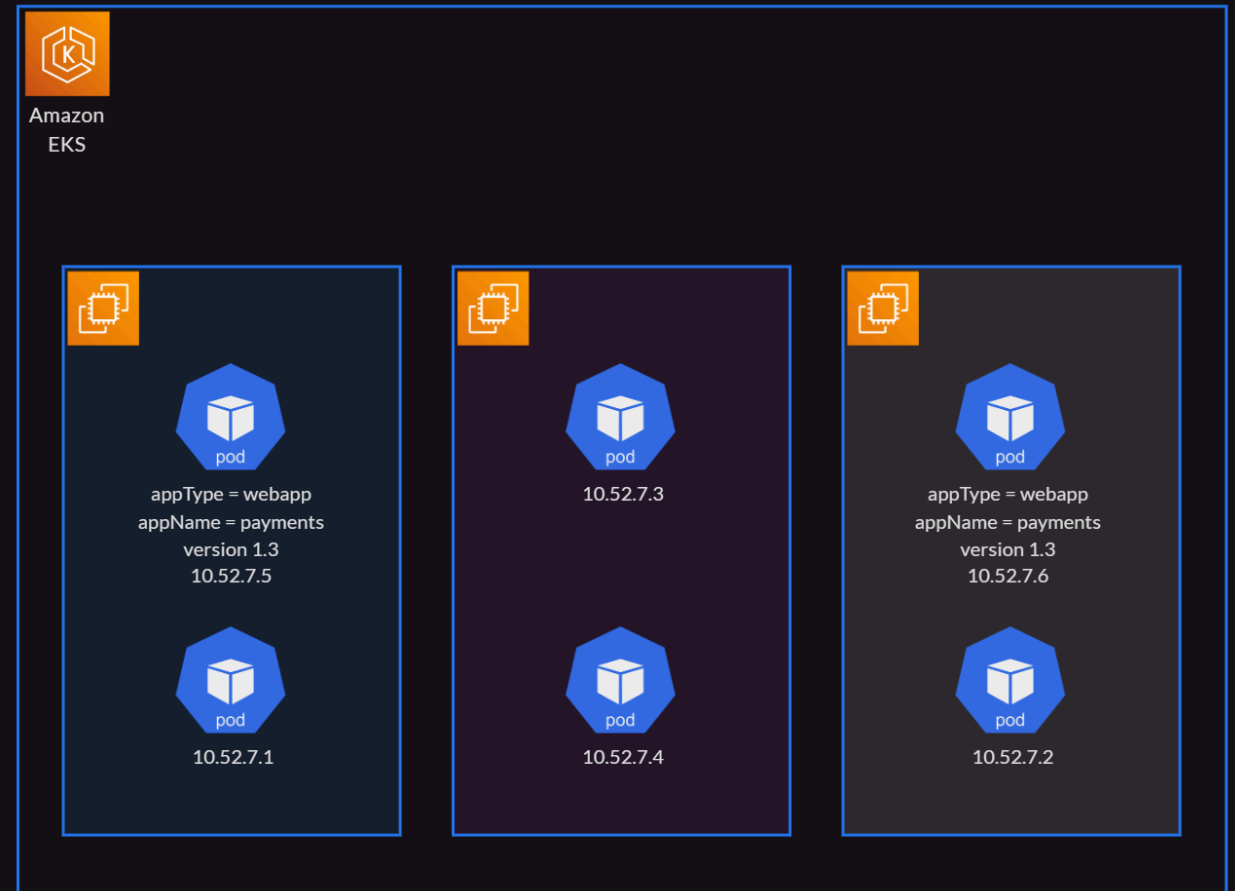


Networking in Kubernetes

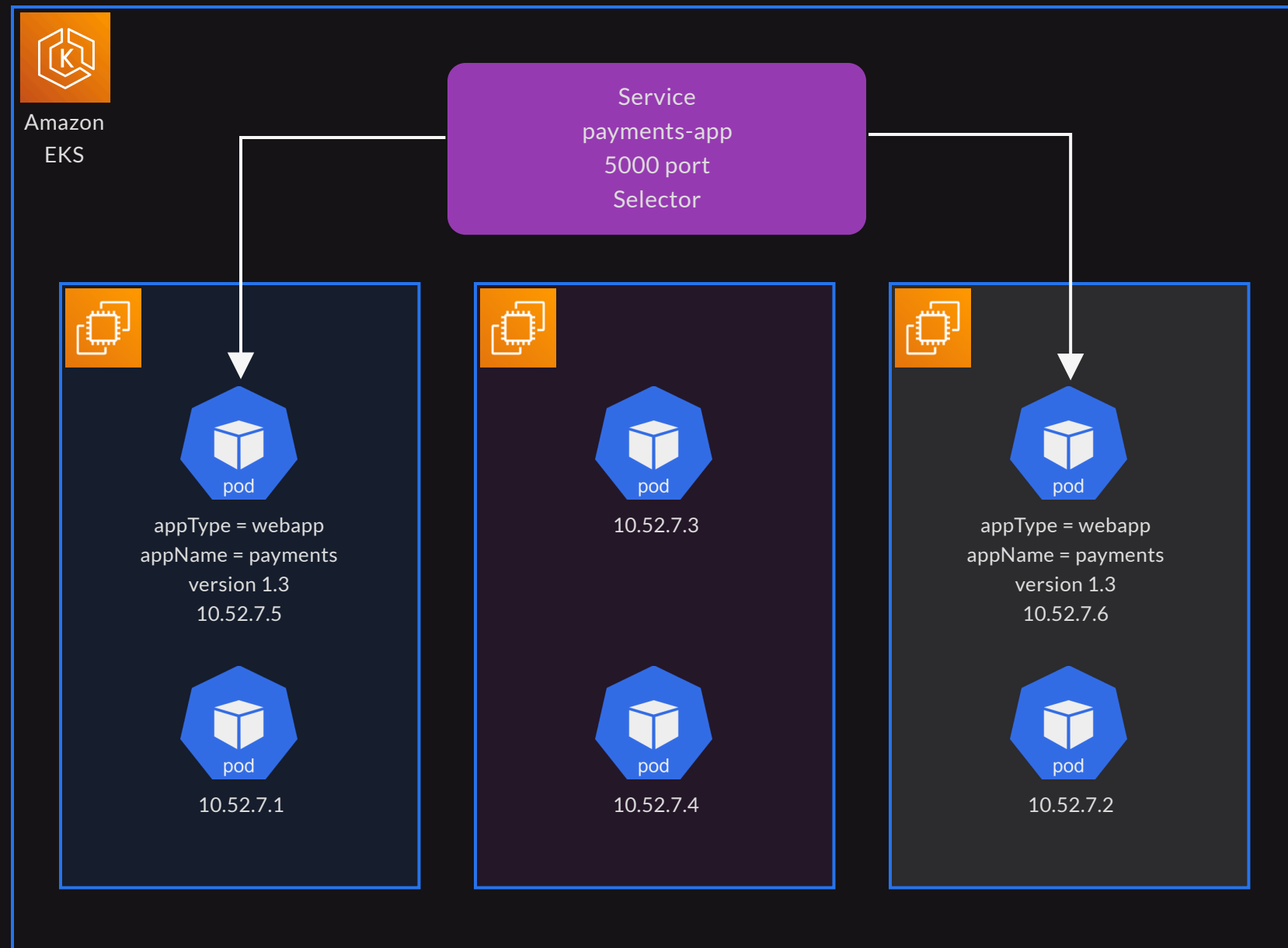


Networking in Kubernetes

- Each pod gets its IP address by the internal CNI (container networking interface)
- Given our payment app, how will the user connect to this app?
- What happens if the node dies leading to a pod created on another node? How will the communication happen?

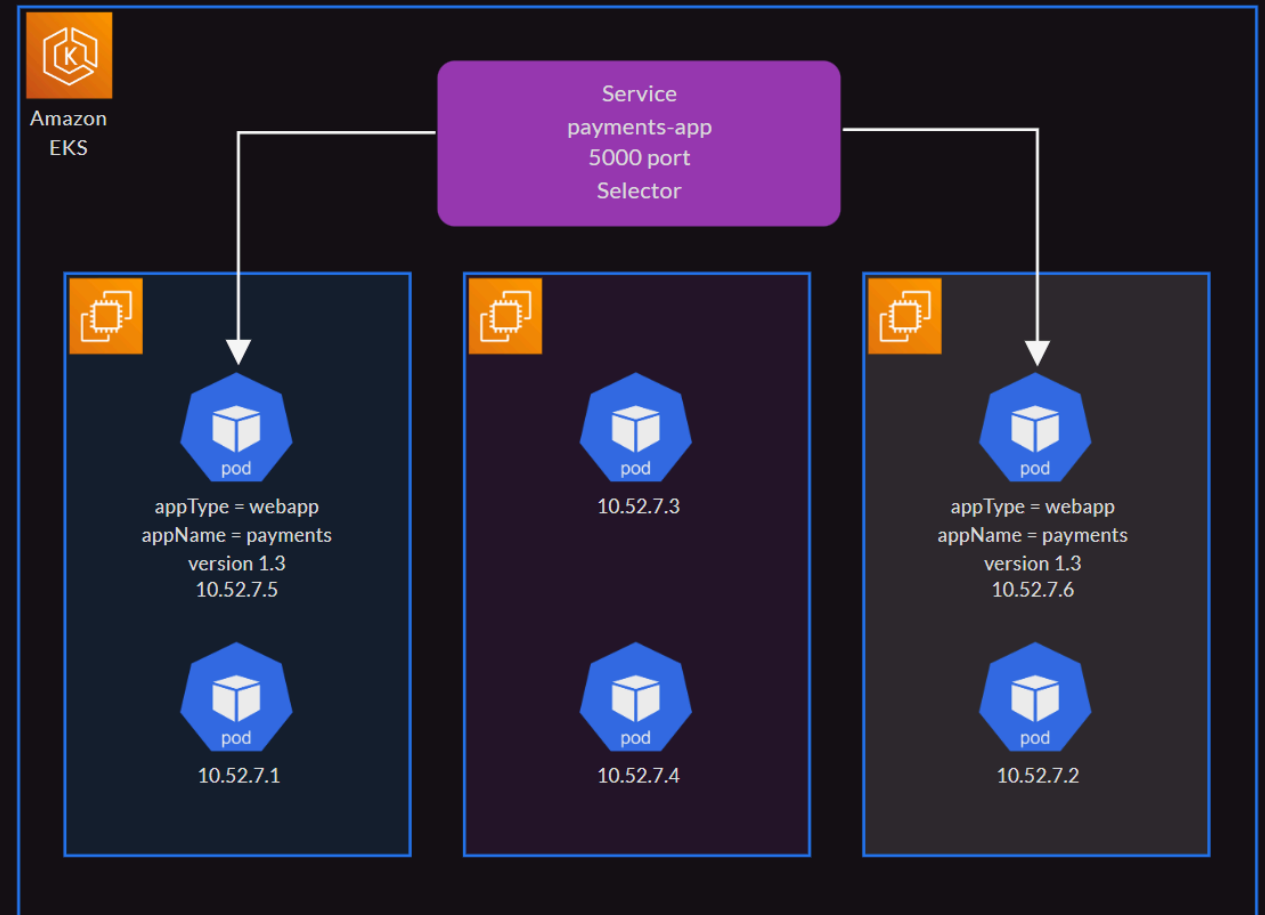


Service Type



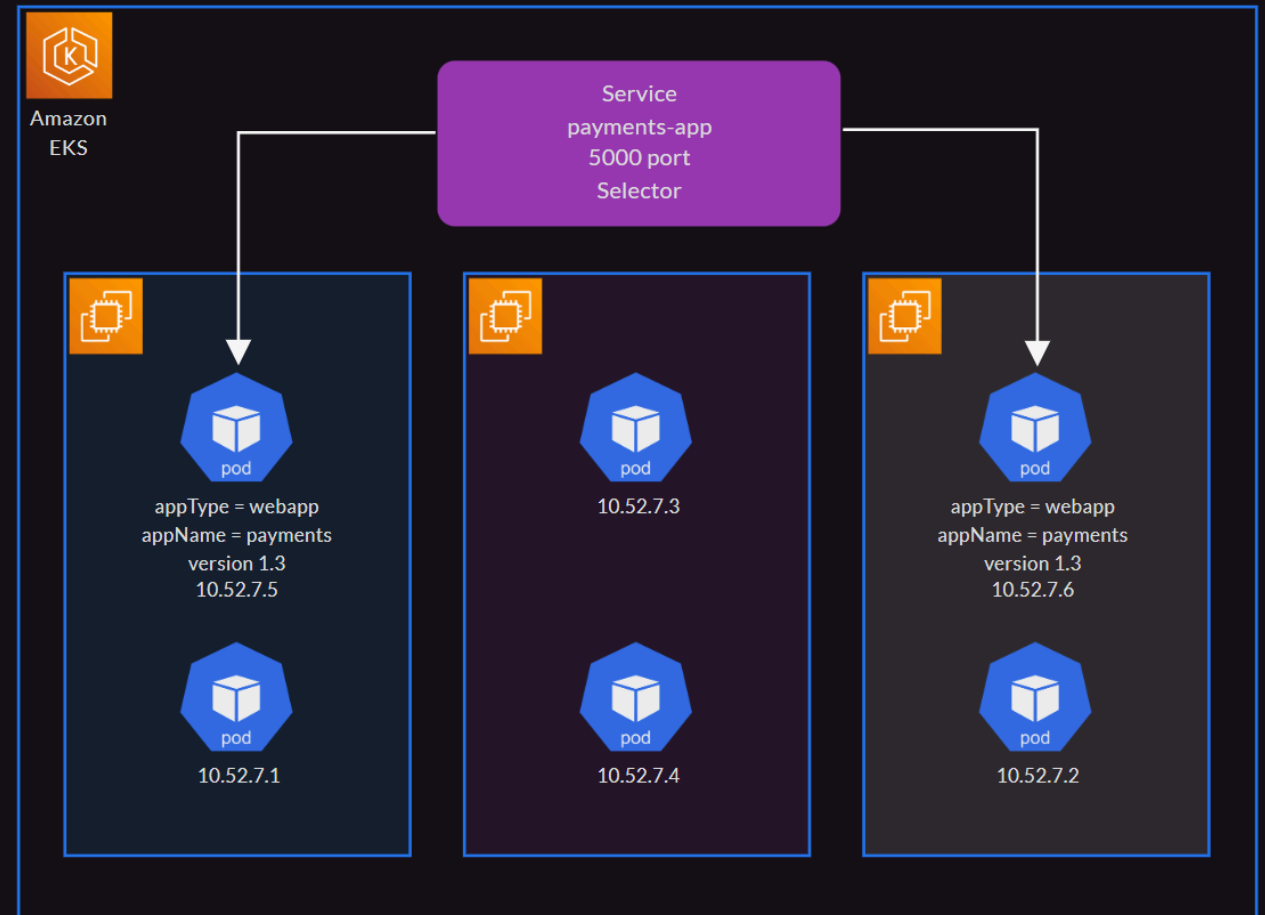
Service Type

- A Kubernetes Service is an abstraction that defines a logical set of Pods running within your cluster, all providing the same functionality
- When created, each Service is assigned a unique IP address (also called ClusterIP) which doesn't change and is added to the internal DNS
- Service uses selectors to find its pod of the same type



Service Type

- Communication to the Service will be automatically load-balanced to some pod that is a member of the Service
- In this case, a microservice can connect to the payments app using <http://payments-app:5000> where 5000 is the service port which will hit target port 8000 of the pods



Thank You
